

Homework 7: Review 2

1.a.

```
String input = in.nextLine();  
  
String upperLetters = "";  
  
for (int i = 0; i < input.length(); i++){  
    char c = input.charAt(i);  
  
    if (Character.isUpperCase(c)){  
        upperLetters += c;  
    }  
}  
  
out.println(upperLetters);
```

b.

```
String input = in.nextLine();  
  
String letter = "";  
  
for (int i = 1; i < input.length(); i += 2){  
    letter += input.charAt(i);  
}  
  
out.println(letter);
```

c.

```
String input = in.nextLine();  
  
String replace = "";  
  
for (int i = 0; i < input.length(); i++){
```

```

        char c = input.charAt(i);

        if (c == 'A' || c == 'E' || c == 'I' || c == 'O' || c == 'U' || c == 'a' || c == 'e'
|| c == 'i' || c == 'o' || c == 'u'){

            replace += '-';

        }else{

            Replace += c;

        }

    }

    out.println(replace);

```

d.

```

String input = in.nextLine();

int count = 0;

for (int i = 0; i < input.length(); i++){

    char c = input.charAt(i);

    if (c == 'A' || c == 'E' || c == 'I' || c == 'O' || c == 'U' || c == 'a' || c == 'e'
|| c == 'i' || c == 'o' || c == 'u'){

        count ++;

    }

}

Out.println(count);

```

e.

```

String input = in.nextLine();

```

```
String position = "";
for (int i = 0; i < input.length(); i++){
    char c = input.charAt(i);
    if (c == 'A' || c == 'E' || c == 'I' || c == 'O' || c == 'U' || c == 'a' || c == 'e'
|| c == 'i' || c == 'o' || c == 'u'){
        position += i;
    }
}
out.println(position);
```

2.

- a. {1, 1, 1, 1, 1, 1, 1, 1, 1, 1}
- b. {1, 1, 2, 3, 4, 5, 4, 3, 2, 1}
- c. {2, 3, 4, 5, 4, 3, 2, 1, 0, 0}
- d. {0, 0, 0, 0, 0, 0, 0, 0, 0, 0}
- e. {1, 3, 6, 10, 15, 19, 22, 24, 25, 25}
- f. {1, 0, 3, 0, 5, 0, 3, 0, 1, 0}
- g. {1, 2, 3, 4, 5, 1, 2, 3, 4, 5}
- h. {1, 1, 2, 3, 4, 4, 3, 2, 1, 0}

3.

```
int max = a [0];
int min = a [0];
for (int i = 1; i < a.length(); i++){
```

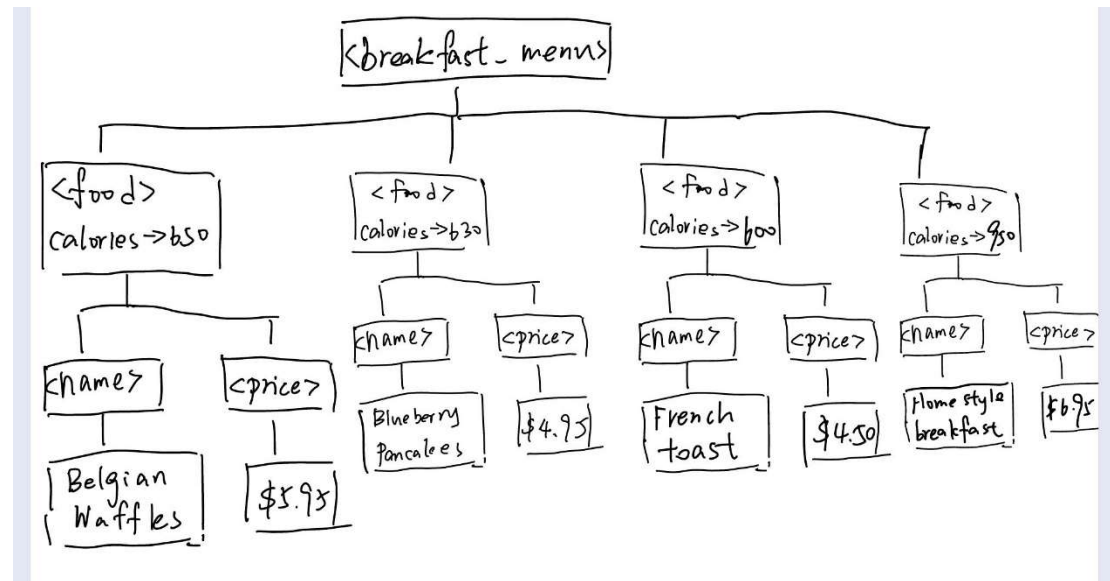
```
    if (a [i] > max){  
        max = a [i];  
    }  
    if (a [i] < min){  
        min = a [i];  
    }  
}
```

4.

```
boolean isOrdered = true;  
for (int i = 0; i < a.length(); i++){  
    if ( a[i] > a[i+1] ){  
        isOrdered = false;  
    }  
}
```

5.

a.



b.

1. true

2. breakfast.menu

3. 4

4. false

5. <food calories= "630">

6. true

7. food

8. 2

9. true

10. 630

11. true;

12. false

13. Blueberry Pancakes

c.

```
<breakfast_menu>

  <food calories="650">

    <name price="$5.95">Belgian Waffles</name>

  </food>

  <food calories="630">

    <name price="$4.95">Blueberry Pancakes</name>

  </food>

  <food calories="600">

    <name price="$4.50">French Toast</name>

  </food>

  <food calories="950">

    <name price="$6.95">Homestyle Breakfast</name>

  </food>

</breakfast_menu>
```

d.

